

Chenmu Zhang

Houston, TX

Mobile: (+1) 737-228-8750 Email: zcmben2014@gmail.com

chenmuzhang.com Google Scholar github.com/cz2014

RESEARCH INTERESTS

- **AI for materials discovery:** LLM agents and machine learning for first-principles simulations
- **Transport in low-dimensional materials:** carrier mobility, thermal transport, and defect dynamics

EDUCATION

- **University of Texas at Austin** Aug. 2018 – Apr. 2023
Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: “*Phonon-limited Carrier Transport in 2D Materials from First Principles*”
- **University of Science and Technology of China (USTC)** Sept. 2014 – Jul. 2018
B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: “*First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain*”

RESEARCH EXPERIENCE

- **Postdoctoral Research Associate** May 2023 – Present
Advisor: Prof. Boris I. Yakobson, Rice University, Houston, TX, United States
Built autonomous LLM agents (MatClaw) that orchestrate multi-code first-principles workflows on HPC clusters for end-to-end materials exploration
Defect physics as a design lever: stochastic defect dynamics in 2D semiconductors as an entropy source for true random bits; mechanisms of defect-governed heat transport at extreme temperatures
- **Graduate Research Assistant** Aug. 2018 – Apr. 2023
Advisor: Prof. Yuanyue Liu, University of Texas at Austin, Austin, TX, United States
Established why 2D semiconductors generally have low mobility and which electronic structures escape it, guiding the discovery of room-temperature high-mobility 2D semiconductors (PRL 2020, 2023)
Extended first-principles electron-phonon transport to quadrupole, substrate, surface, and remote-phonon scattering, revealing anomalous mobility enhancement by remote phonons (Nano Lett. 2026); built a computational carrier-mobility genome for 2D semiconductors

PUBLICATIONS

17 publications (12 first-author), including *Phys. Rev. Lett.*, *J. Am. Chem. Soc.*, *ACS Nano*, and *Nano Lett.*; 613 citations, h-index 10 (Google Scholar, Aug. 2026).

17. **Zhang***, C.; Felix*, L. C.; Yakobson, B. I. Defect-controlled thermal conductivity of ZrC at high temperatures from first principles. (Submitted) **2026**.
16. **Zhang**, C.; Sun, F.; Yakobson, B. I. Design of efficient true random bits generation in transition metal dichalcogenides from first principles. (Submitted) **2026**.
15. **Zhang**, C.; Yakobson, B. I. Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop. arXiv preprint arXiv:2606.29717, accepted at the Workshop on Language Models for Scientific Discovery (LM4Sci), COLM **2026**.
14. **Zhang**, C.; Liu, Y. Enhancing Carrier Mobility by Remote Phonons. *Nano Letters* **2026**, 26(30), 9716–9721.
13. **Zhang**, C.; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688 **2026**.

12. **Zhang, C.**; Liu, Y. Two-dimensional Semiconductor Computational Carrier Mobility Genome. arXiv preprint arXiv:2404.08117 **2024**.
11. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439.
10. **Zhang, C.**; Xiao, Z.; Paddock, R.; Cullinan, M.; Tehrani, M.; Liu, Y. Effects of Graphene Doping on the Electrical Conductivity of Copper. *Advanced Functional Materials* **2024**, 34(45), 2407569.
9. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001.
8. **Zhang, C.**; Liu, Y. Phonon-limited transport of two-dimensional semiconductors: Quadrupole scattering and free carrier screening. *Physical Review B* **2022**, 106(11), 115423.
7. **Zhang, C.**; Cheng, L.; Liu, Y. Understanding high-field electron transport properties and strain effects of monolayer transition metal dichalcogenides. *Physical Review B* **2020**, 102(11), 115405.
6. **Zhang, C.**; Cheng, L.; Liu, Y. Role of flexural phonons in carrier mobility of two-dimensional semiconductors: free standing vs on substrate. *Journal of Physics: Condensed Matter* **2021**, 33(23), 234003.
5. Xiao, Z.; Guo, R.; **Zhang, C.**; Liu, Y. Point Defect Limited Carrier Mobility in 2D Transition Metal Dichalcogenides. *ACS Nano* **2024**, 18(11), 8511–8516.
4. Cheng, L.; **Zhang, C.**; Liu, Y. Intrinsic charge carrier mobility of 2D semiconductors. *Computational Materials Science* **2021**, 194, 110468.
3. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701.
2. Cheng, L.; **Zhang, C.**; Liu, Y. The Optimal Electronic Structure for High-Mobility 2D Semiconductors: Exceptionally High Hole Mobility in 2D Antimony. *Journal of the American Chemical Society* **2019**, 141(41), 16296–16302.
1. Cheng, L.; **Zhang, C.**; Liu, Y. How to resolve a phonon-associated property into contributions of basic phonon modes. *Journal of Physics: Materials* **2019**, 2(4), 045005.

SOFTWARE

- **MatClaw** — autonomous LLM agent for end-to-end first-principles materials research. Sole developer.
- **qeh-frohlich** — phonon-limited mobility in van der Waals heterostructures with Fröhlich coupling. Sole developer.
- **EDI** — first-principles electron–defect scattering and defect-limited transport. Contributor.

PRESENTATIONS

- *Autonomous Materials Simulation with an LLM Agent: Modeling Exothermic Fronts in 2D Ferroelectrics*, APS Global Physics Summit, Denver, 2026 (talk)
- *First-Principles Electron-Surface Scattering: Case Study on Copper Films*, APS Global Physics Summit, Anaheim, 2025 (talk); *Harnessing Stochasticity in 2D Materials: Phonon-Assisted Defect Charge Dynamics in WS₂*, same meeting (poster)
- *Electron-Surface Scattering from First Principles*, APS March Meeting, Minneapolis, 2024 (talk); *New 2D Semiconductors with Ultra-High Mobility at Room Temperature*, same meeting (poster)

PROFESSIONAL SERVICE

- Reviewer for *Physical Review B*, *Ceramics International*, and *Journal of Superconductivity and Novel Magnetism*

TEACHING

- Teaching Assistant, Structure of Materials (Fall 2021, Fall 2022), University of Texas at Austin